

Reduced-Order Slosh Modeling via Moving Horizon Parameter Identification

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Abstract

Accurate prediction of slosh-induced forces in partially filled propellant tanks remains a critical challenge for guidance, navigation, and control (GNC) applications, particularly when high-fidelity fluid effects must be retained at a manageable computational expense. To address this, the present work develops a reduced-order slosh estimation framework for a rectangular tank subjected to prescribed lateral excitation. The methodology is grounded in high-fidelity computational fluid dynamics (CFD) simulations using OpenFOAM, which characterize the liquid center-of-mass motion and wall reaction forces across multiple fill levels. These simulated outputs serve as the primary measurement inputs for a two-mode reduced-order model designed to capture dominant lateral slosh dynamics. The objective of the estimation is to identify both the internal slosh states and the governing model parameters required to accurately reproduce the CFD-derived force behavior.

To extend the framework beyond discrete simulated conditions, the time-varying parameter sequences produced by the proposed estimator are synthesized into analytical functions for each fuel height. These functions are systematically integrated to enforce continuity across the fill-level domain, transforming isolated estimates into smooth, multidimensional parameter fields. By leveraging the time-varying nature of the estimates derived in this study, the resulting parameterizations allow the reduced-order model to reconstruct slosh-induced force responses across a continuous envelope. This ensures that the model remains both physically consistent and computationally compact, transitioning seamlessly between varying operating points across the entire flight envelope.

The estimation framework is formulated using Moving Horizon Estimation (MHE), derived from a finite-horizon maximum a posteriori problem. This formulation yields a covariance-weighted least-squares cost function that balances measurement data, arrival costs, and dynamic-defect terms. To efficiently transition between successive time horizons, the algorithm uses a local second-order approximation of the cost function; the curvature (Hessian) of this approximation defines the posterior uncertainty, allowing for a recursive update of the arrival covariance. Overall, this approach provides a consistent framework that integrates high-fidelity CFD data with recursive state and parameter estimation. The resulting model enables efficient reconstruction and prediction of slosh-induced forces, making it well-suited for integration into GNC simulation environments.

I. INTRODUCTION

We consider a partially filled rectangular tank subjected to a prescribed lateral acceleration $a_x(t)$. The resulting fluid motion generates pressure distributions, velocity fields, and wall stresses; however, for estimation and control, the most relevant quantities are the liquid center-of-mass displacement and the net hydrodynamic force on the tank walls. In this work, these quantities are extracted from high-fidelity CFD simulations [1] and used to identify a reduced-order slosh model within a moving horizon estimation framework [2], which naturally accommodates the state and parameter constraints inherent to the problem. Although the CFD model resolves the fluid dynamics in full detail, it is computationally prohibitive for recursive estimation, often requiring hours of processing to simulate only seconds of physical time. A reduced-order model is therefore introduced to retain the dominant slosh dynamics while remaining low-dimensional enough for real-time implementation, enabling the orders-of-magnitude speedup required for GNC applications.

To achieve this, the fluid motion is approximated using a two-mode reduced-order model that captures the dominant lateral dynamics, as higher-order modes primarily introduce only short-lived transients [3]. Because the applied forcing dictates a predominantly lateral response, the formulation is restricted strictly to the excitation axis, which inherently contains the principal information needed to reconstruct both the center-of-mass motion and wall forces. The resulting model couples these modal states with a parametric force representation, requiring the joint estimation of twelve time-varying parameters. While these parameters depend on specific tank geometry, the underlying modeling and estimation framework readily generalizes to other container shapes and excitation conditions.

All code required to reproduce the results is available at an open-source GitHub repository [5].

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The measurements used by the estimator are

$$y_k = \begin{bmatrix} q_{\text{meas},k} \\ F_{x,\text{meas},k} \end{bmatrix}, \quad (1)$$

where $q_{\text{meas},k}$ is the liquid center-of-mass displacement in the excitation direction and $F_{x,\text{meas},k}$ is the net hydrodynamic force on the tank walls in the same direction.

To represent the dominant slosh response, the fluid motion is approximated by a two-mode reduced-order model,

$$\ddot{q}_1 + 2\zeta_1\omega_1\dot{q}_1 + \omega_1^2 q_1 = \gamma_1 a_x(t), \quad \ddot{q}_2 + 2\zeta_2\omega_2\dot{q}_2 + \omega_2^2 q_2 = \gamma_2 a_x(t), \quad (2)$$

with total displacement

$$q = q_1 + q_2. \quad (3)$$

The measured wall force is modeled as

$$F_x = c_1 q_1 + c_2 \dot{q}_1 + c_3 q_2 + c_4 \dot{q}_2 + d a_x + b. \quad (4)$$

The modal and force-model parameters are not assumed known a priori and are therefore estimated jointly with the modal states.

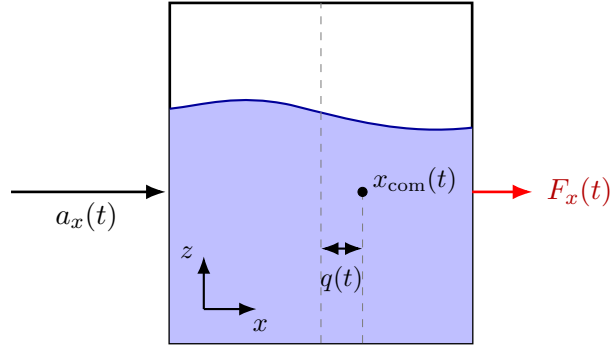


Fig. 1: Rectangular tank subjected to lateral acceleration $a_x(t)$. The measured quantities are the center-of-mass displacement $q(t)$ relative to the tank centerline and the hydrodynamic wall force $F_x(t)$.

II. PHYSICAL SYSTEM AND REDUCED-ORDER MODEL

A. Measurement interpretation and direction choice

Figure 1 summarizes the physical quantities used by the estimator. The quantity $q_{\text{meas},k}$ measures bulk fluid redistribution in the excitation direction, while $F_{x,\text{meas},k}$ measures the dynamic load transmitted to the tank walls. These signals are complementary: the center-of-mass displacement provides kinematic information, while the wall force provides dynamic information.

The present formulation is restricted to the x -direction because the imposed excitation is lateral and the dominant measured response is also lateral. A multidirectional model would increase estimator dimension and parameter ambiguity without comparable observability in the current dataset. The one-direction model is therefore a physically motivated reduction rather than an arbitrary simplification.

B. Two-mode model and force parametrization

The fluid motion is approximated by two dominant slosh modes,

$$\ddot{q}_1 + 2\zeta_1\omega_1\dot{q}_1 + \omega_1^2 q_1 = \gamma_1 a_x(t), \quad (5)$$

$$\ddot{q}_2 + 2\zeta_2\omega_2\dot{q}_2 + \omega_2^2 q_2 = \gamma_2 a_x(t), \quad (6)$$

with total displacement

$$q = q_1 + q_2. \quad (7)$$

The wall force is modeled by

$$F_x = c_1 q_1 + c_2 \dot{q}_1 + c_3 q_2 + c_4 \dot{q}_2 + d a_x + b. \quad (8)$$

The coefficients c_1 and c_3 capture displacement-dependent contributions, c_2 and c_4 capture velocity-dependent contributions, d accounts for direct inertial coupling with the imposed acceleration, and b absorbs constant bias and residual model mismatch.

Two slosh modes are used because a single oscillator is generally insufficient to reproduce both the low-frequency bulk motion and the higher-frequency oscillatory content observed in the CFD data. The two-mode representation therefore provides the smallest model capable of reproducing the principal slosh features while remaining tractable for estimation.

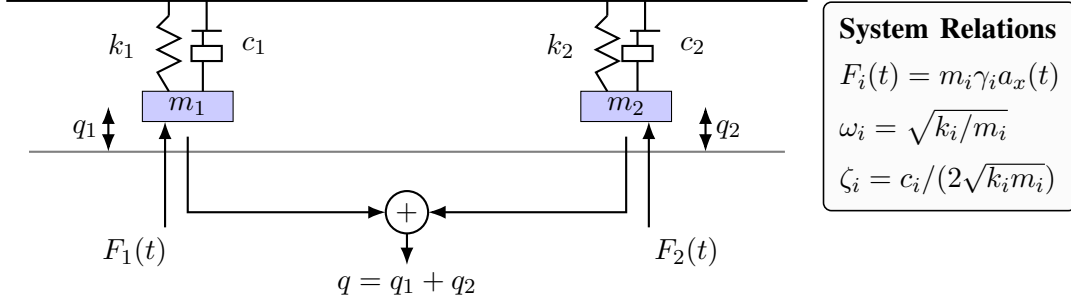


Fig. 2: Mechanical analogy for the two-mode slosh model.

C. State definition and parameter augmentation

Define the physical state as

$$x_k = [q_{1,k} \quad \dot{q}_{1,k} \quad q_{2,k} \quad \dot{q}_{2,k}]^T \in \mathbb{R}^4, \quad (9)$$

and define the parameter vector as

$$\theta_k = [\omega_{1,k} \quad \zeta_{1,k} \quad \gamma_{1,k} \quad \omega_{2,k} \quad \zeta_{2,k} \quad \gamma_{2,k} \quad c_{1,k} \quad c_{2,k} \quad c_{3,k} \quad c_{4,k} \quad d_k \quad b_k]^T \in \mathbb{R}^{12}. \quad (10)$$

The augmented state is

$$X_k = \begin{bmatrix} x_k \\ \theta_k \end{bmatrix} \in \mathbb{R}^{16}. \quad (11)$$

This augmented formulation allows the estimator to adjust both modal states and reduced-order parameters online.

Thus, the estimation problem involves recovering a 16-dimensional augmented state from only two measured quantities. Specifically, the measurement vector y_k provides the center-of-mass displacement and wall force, while the excitation input $a_x(t)$ is assumed known. In contrast, both the modal states x_k and the parameter vector θ_k are treated as unknown and must be estimated online.

This places strong demands on system observability. In this setting, parameter identifiability depends critically on the excitation input $a_x(t)$, which must provide sufficient richness, commonly referred to as persistence of excitation, to ensure local observability of the augmented state over the estimation horizon. The broadband frequency content of the CFD-prescribed excitation ensures that both modes are sufficiently stimulated, enabling the estimator to distinguish among parameters governing stiffness (ω), damping (ζ), and forcing gains (γ) [3].

III. NONLINEAR STOCHASTIC STATE-SPACE MODEL

A. Discrete-time nonlinear dynamics and measurement model

Using forward Euler discretization with step size Δt_k , selected for its simplicity, consistency with the recursive estimation framework, and straightforward Jacobian evaluation while remaining sufficiently accurate for small time steps, the modal dynamics become

$$q_{1,k+1} = q_{1,k} + \Delta t_k \dot{q}_{1,k}, \quad (12)$$

$$\dot{q}_{1,k+1} = \dot{q}_{1,k} + \Delta t_k (-2\zeta_{1,k}\omega_{1,k}\dot{q}_{1,k} - \omega_{1,k}^2 q_{1,k} + \gamma_{1,k} a_{x,k}), \quad (13)$$

$$q_{2,k+1} = q_{2,k} + \Delta t_k \dot{q}_{2,k}, \quad (14)$$

$$\dot{q}_{2,k+1} = \dot{q}_{2,k} + \Delta t_k (-2\zeta_{2,k}\omega_{2,k}\dot{q}_{2,k} - \omega_{2,k}^2 q_{2,k} + \gamma_{2,k} a_{x,k}). \quad (15)$$

For the present fixed-fill experiments, the parameters are modeled as slow random walks,

$$\theta_{k+1} = \theta_k + \eta_k. \quad (16)$$

where η_k represents zero-mean process noise. This models slow parameter variations arising from reduced-order approximation error and unmodeled dynamics, and allows the estimator to adapt the parameters over time.

The resulting augmented-state model is nonlinear due to the multiplicative coupling between states and parameters, although each modal equation is linear for fixed parameter values.

The corresponding augmented nonlinear dynamics can be written compactly as

$$f(X_k, u_k) = \begin{bmatrix} q_{1,k} + \Delta t_k \dot{q}_{1,k} \\ \dot{q}_{1,k} + \Delta t_k \left(-2\zeta_{1,k}\omega_{1,k}\dot{q}_{1,k} - \omega_{1,k}^2 q_{1,k} + \gamma_{1,k} a_{x,k} \right) \\ q_{2,k} + \Delta t_k \dot{q}_{2,k} \\ \dot{q}_{2,k} + \Delta t_k \left(-2\zeta_{2,k}\omega_{2,k}\dot{q}_{2,k} - \omega_{2,k}^2 q_{2,k} + \gamma_{2,k} a_{x,k} \right) \\ \theta_k \end{bmatrix}. \quad (17)$$

The measurement model is defined as

$$y_k = h(X_k, u_k) + G_k v_k, \quad (18)$$

with

$$h(X_k, u_k) = \begin{bmatrix} q_{1,k} + q_{2,k} \\ c_{1,k}q_{1,k} + c_{2,k}\dot{q}_{1,k} + c_{3,k}q_{2,k} + c_{4,k}\dot{q}_{2,k} + d_k a_{x,k} + b_k \end{bmatrix}. \quad (19)$$

The first output predicts the center-of-mass displacement and the second predicts the wall force.

B. Stochastic embedding for estimation

For estimation, the nonlinear system is written in stochastic state-space form as

$$X_{k+1} = f(X_k, u_k) + C_k W_k, \quad (20)$$

$$y_k = h(X_k, u_k) + G_k v_k. \quad (21)$$

where:

$$C_k = I_{n_x}, \quad W_k \sim \mathcal{N}(0, Q_k^X), \quad (22)$$

$$G_k = I_{n_y}, \quad v_k \sim \mathcal{N}(0, R_k). \quad (23)$$

Here W_k denotes the augmented state noise, modeled as zero-mean white Gaussian noise with covariance Q_k^X , and v_k denotes the measurement noise, modeled as zero-mean white Gaussian noise with covariance R_k [4]. The process noise accounts for reduced-order model mismatch, neglected higher-order fluid dynamics, and parameter drift, while the measurement noise represents uncertainty in the CFD-derived quantities.

The estimator is initialized with

$$X_0 \sim \mathcal{N}(\hat{X}_0^{(-)}, P_0^{(-)}), \quad (24)$$

where $\hat{X}_0^{(-)}$ and $P_0^{(-)}$ denote the initial state estimate (arrival mean) and its associated covariance, specified based on prior knowledge of the system. They serve to propagate prior estimate from the previous estimation window.

To obtain the linear time-varying form used in the MHE cost and covariance propagation, the nonlinear model is locally linearized about the current estimate $\hat{X}_k^{(+)}$:

$$X_{k+1} \approx A_k X_k + B_k u_k + C_k w_k, \quad (25)$$

$$y_k \approx H_k X_k + G_k v_k. \quad (26)$$

Here A_k , H_k , C_k , and G_k are the Jacobians evaluated at the current estimate.

IV. FINITE-HORIZON MAP DERIVATION OF MHE

At time k , define the trailing horizon $[k-N, \dots, k]$. The unknown is the trajectory $X_{k-N:k} := \{X_{k-N}, X_{k-N+1}, \dots, X_k\}$, and the available measurements are $y_{k-N:k} := \{y_{k-N}, y_{k-N+1}, \dots, y_k\}$.

The MHE problem is posed as the finite-horizon MAP estimator [2]

$$\hat{X}_{k-N:k}^{(+)} = \arg \max_{X_{k-N:k}} p(X_{k-N:k} \mid y_{k-N:k}) \quad (27)$$

By Bayes' rule [2],

$$p(X_{k-N:k} \mid y_{k-N:k}) = \frac{p(y_{k-N:k} \mid X_{k-N:k}) p(X_{k-N:k})}{p(y_{k-N:k})}. \quad (28)$$

Since $p(y_{k-N:k})$ depends only on known data and not on the optimization variable $X_{k-N:k}$, it may be dropped for optimization:

$$p(X_{k-N:k} \mid y_{k-N:k}) \propto p(y_{k-N:k} \mid X_{k-N:k}) p(X_{k-N:k}). \quad (29)$$

Using the usual conditional independence assumption for measurements,

$$p(y_{k-N:k} \mid X_{k-N:k}) = \prod_{i=k-N}^k p(y_i \mid X_i), \quad (30)$$

and the Markov assumption for the state trajectory,

$$p(X_{k-N:k}) = p(X_{k-N}) \prod_{i=k-N}^{k-1} p(X_{i+1} \mid X_i). \quad (31)$$

The Markov assumption states that, given the current state X_i , the next state X_{i+1} is conditionally independent of all prior states. This implies that the current state provides a complete and sufficient representation of the system, enabling the reduced-order dynamics to be propagated forward without dependence on past states.

Substituting these factorizations into the posterior gives

$$p(X_{k-N:k} \mid y_{k-N:k}) \propto \left[\prod_{i=k-N}^k p(y_i \mid X_i) \right] \left[p(X_{k-N}) \prod_{i=k-N}^{k-1} p(X_{i+1} \mid X_i) \right]. \quad (32)$$

Taking the negative logarithm converts the product of densities into a sum:

$$\hat{X}_{k-N:k}^{MHE} = \arg \min_{X_{k-N:k}} \left[- \sum_{i=k-N}^k \log p(y_i \mid X_i) - \log p(X_{k-N}) - \sum_{i=k-N}^{k-1} \log p(X_{i+1} \mid X_i) \right]. \quad (33)$$

V. GAUSSIAN DENSITIES AND COVARIANCE-WEIGHTED COST

Over the horizon, consider the locally linearized stochastic model

$$X_{i+1} = A_i X_i + B_i u_i + C_i w_i, \quad w_i \sim \mathcal{N}(0, Q_i), \quad (34)$$

$$y_i = H_i X_i + G_i v_i, \quad v_i \sim \mathcal{N}(0, R_i). \quad (35)$$

The arrival prior at the beginning of the window is

$$X_{k-N} \sim \mathcal{N}(\hat{X}_{k-N}^{(-)}, P_{k-N}^{(-)}). \quad (36)$$

Using standard properties of Gaussian distributions under affine transformations [4],

$$y_i \mid X_i \sim \mathcal{N}(H_i X_i, G_i R_i G_i^T), \quad (37)$$

$$X_{i+1} \mid X_i \sim \mathcal{N}(A_i X_i + B_i u_i, C_i Q_i C_i^T). \quad (38)$$

These expressions follow from the fact that the noise variables $w_i \sim \mathcal{N}(0, Q_i)$ and $v_i \sim \mathcal{N}(0, R_i)$ enter the process and measurement models through affine transformations. Specifically, $y_i = H_i X_i + G_i v_i$ and $X_{i+1} = A_i X_i + B_i u_i + C_i w_i$ are linear functions of the Gaussian noise terms with deterministic offsets. As a result, $y_i \mid X_i$

and $X_{i+1} | X_i$ are themselves Gaussian, with means given by the deterministic components $H_i X_i$ and $A_i X_i + B_i u_i$, and covariances obtained by propagating R_i and Q_i through G_i and C_i , respectively.

Therefore,

$$p(y_i | X_i) = \frac{1}{\sqrt{(2\pi)^{n_y} \det(G_i R_i G_i^T)}} \exp \left[-\frac{1}{2} (y_i - H_i X_i)^T (G_i R_i G_i^T)^{-1} (y_i - H_i X_i) \right], \quad (39)$$

$$p(X_{k-N}) = \frac{1}{\sqrt{(2\pi)^{n_x} \det P_{k-N}^{(-)}}} \exp \left[-\frac{1}{2} (X_{k-N} - \hat{X}_{k-N}^{(-)})^T (P_{k-N}^{(-)})^{-1} (X_{k-N} - \hat{X}_{k-N}^{(-)}) \right], \quad (40)$$

and

$$p(X_{i+1} | X_i) = \frac{1}{\sqrt{(2\pi)^{n_x} \det(C_i Q_i C_i^T)}} \exp \left[-\frac{1}{2} (X_{i+1} - A_i X_i - B_i u_i)^T (C_i Q_i C_i^T)^{-1} (X_{i+1} - A_i X_i - B_i u_i) \right]. \quad (41)$$

Substituting these densities into (33) and dropping normalization constants that do not depend on the optimization variable yields the MHE cost

$$\begin{aligned} J_{k-N:k} = & \sum_{i=k-N}^k (y_i - H_i X_i)^T (G_i R_i G_i^T)^{-1} (y_i - H_i X_i) \\ & + (X_{k-N} - \hat{X}_{k-N}^{(-)})^T (P_{k-N}^{(-)})^{-1} (X_{k-N} - \hat{X}_{k-N}^{(-)}) \\ & + \sum_{i=k-N}^{k-1} (X_{i+1} - A_i X_i - B_i u_i)^T (C_i Q_i C_i^T)^{-1} (X_{i+1} - A_i X_i - B_i u_i). \end{aligned} \quad (42)$$

This contains three components: a measurement-fit term, an arrival term, and a dynamic-consistency term.

The arrival term represents the prior information at the beginning of the estimation window. It summarizes all past measurements prior to time $k - N$ through the arrival mean $\hat{X}_{k-N}^{(-)}$ and covariance $P_{k-N}^{(-)}$. In this way, the MHE formulation maintains recursive consistency across successive horizons. Rather than reinitializing the estimator at each window shift, the arrival term propagates information forward, playing a role analogous to the prediction step in a Kalman filter and providing the necessary memory for parameter convergence [4].

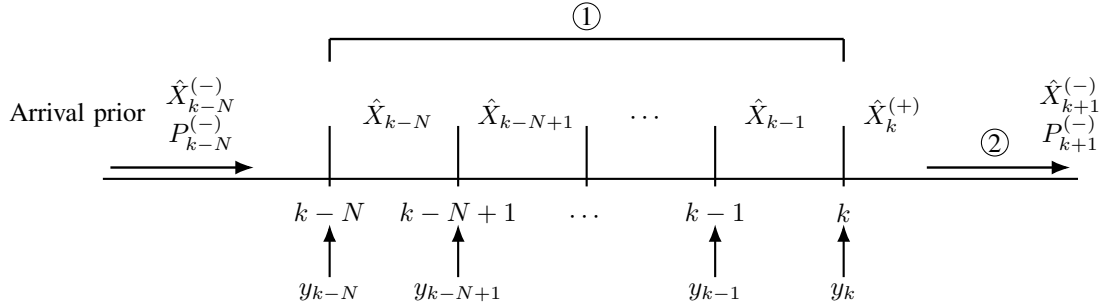


Fig. 3: Moving Horizon Estimation recursion. At each time step, a batch optimization is performed over the horizon $[k - N, \dots, k]$ using the available measurements. The terminal estimate $(\hat{X}_k^{(+)}, P_k^{(+)})$ is then propagated to form the arrival prior $(\hat{X}_{k+1}^{(-)}, P_{k+1}^{(-)})$ for the next horizon.

VI. RESIDUAL FORM AND LOCAL COVARIANCE APPROXIMATION

Define the stacked decision vector

$$z := [X_{k-N}^T \quad X_{k-N+1}^T \quad \dots \quad X_k^T]^T. \quad (43)$$

Define the residuals

$$e_i^y(z) := y_i - H_i X_i, \quad e^a(z) := X_{k-N} - \hat{X}_{k-N}^{(-)}, \quad e_i^d(z) := X_{i+1} - A_i X_i - B_i u_i. \quad (44)$$

With the covariance-based weights

$$W_i^y := (G_i R_i G_i^T)^{-1}, \quad W^a := (P_{k-N}^{(-)})^{-1}, \quad W_i^d := (C_i Q_i C_i^T)^{-1}, \quad (45)$$

the cost can be written compactly as

$$J_{k-N:k}(z) = \sum_{i=k-N}^k e_i^y(z)^T W_i^y e_i^y(z) + e^a(z)^T W^a e^a(z) + \sum_{i=k-N}^{k-1} e_i^d(z)^T W_i^d e_i^d(z). \quad (46)$$

This form shows that MHE is a weighted least-squares problem over the full horizon.

Let $\hat{z}$, with $z = X_{k-N:k}$, denote the optimizer of the MHE problem. Since $J_{k-N:k}(z)$ is smooth, it admits a second-order Taylor expansion about $\hat{z}$:

$$J_{k-N:k}(z) = J_{k-N:k}(\hat{z}) + \nabla J_{k-N:k}(\hat{z})^T (z - \hat{z}) + \frac{1}{2} (z - \hat{z})^T \mathcal{H}_k (z - \hat{z}) + \mathcal{O}(\|z - \hat{z}\|^3), \quad (47)$$

where

$$\mathcal{H}_k := \left. \frac{\partial^2 J_{k-N:k}}{\partial z^2} \right|_{z=\hat{z}}. \quad (48)$$

At the local minimizer, $\nabla J_{k-N:k}(\hat{z}) = 0$, so near the optimum

$$J_{k-N:k}(z) \approx J_{k-N:k}(\hat{z}) + \frac{1}{2} (z - \hat{z})^T \mathcal{H}_k (z - \hat{z}). \quad (49)$$

The Hessian $\mathcal{H}_k$ characterizes the local curvature of the cost function at the optimizer. A large curvature implies that small deviations from the solution result in a significant increase in cost, corresponding to high confidence (low uncertainty), whereas a shallow curvature indicates greater uncertainty.

This local quadratic structure motivates a Gaussian approximation of the posterior distribution near the optimizer,

$$p(z \mid y_{k-N:k}) \approx \mathcal{N}(\hat{z}, \mathcal{H}_k^{-1}), \quad (50)$$

so that the terminal covariance is approximated by

$$P_k^{(+)} \approx [\mathcal{H}_k^{-1}]_{X_k, X_k}. \quad (51)$$

In nonlinear MHE, this represents a local curvature-based approximation rather than an exact covariance. Starting from the residual form of the cost, the Hessian evaluated at the optimizer is

$$\mathcal{H}_k = \sum_{\ell} \left(\frac{\partial e_{\ell}}{\partial z} \right)^T W_{\ell} \left(\frac{\partial e_{\ell}}{\partial z} \right) + \sum_{\ell} e_{\ell}(z) \frac{\partial^2 e_{\ell}}{\partial z^2}. \quad (52)$$

Near the optimum, the residuals are small and the second term is neglected, yielding the Gauss–Newton approximation

$$\mathcal{H}_k \approx \sum_{\ell} \left(\frac{\partial e_{\ell}}{\partial z} \right)^T W_{\ell} \left(\frac{\partial e_{\ell}}{\partial z} \right) = J^T J, \quad (53)$$

where $J = \partial r / \partial z$ is the Jacobian of the stacked residual vector.

Thus, the terminal covariance used in the estimator is approximated as

$$P_k^{(+)} \approx [(J^T J)^{-1}]_{X_k, X_k}. \quad (54)$$

VII. RECURSIVE ARRIVAL MEAN AND COVARIANCE UPDATE

The MHE recursion follows the two-step structure shown in Fig. 3. After solving the horizon problem, the terminal estimate is propagated forward to form the next arrival prior. Using the locally linearized model,

$$X_{k+1} = A_k X_k + B_k u_k + C_k w_k, \quad (55)$$

the propagated arrival mean is

$$\hat{X}_{k+1}^{(-)} = A_k \hat{X}_k^{(+)} + B_k u_k, \quad (56)$$

and the corresponding arrival covariance is

$$P_{k+1}^{(-)} \approx A_k P_k^{(+)} A_k^T + C_k Q_k C_k^T. \quad (57)$$

This update is Riccati-like in the sense that it mirrors Kalman covariance propagation, but here the posterior covariance $P_k^{(+)}$ is obtained from the local curvature of the finite-horizon optimization problem via a Gauss–Newton approximation [4].

Therefore, the recursive covariance structure used by the estimator is

$$P_k^{(+)} \approx [(J^T J)^{-1}]_{X_k, X_k}, \quad P_{k+1}^{(-)} \approx A_k P_k^{(+)} A_k^T + C_k Q_k C_k^T, \quad (58)$$

together with the state propagation

$$\hat{X}_{k+1}^{(-)} = A_k \hat{X}_k^{(+)} + B_k u_k. \quad (59)$$

Let $r(z)$ denote the stacked residual vector formed by concatenating the arrival, dynamics, and measurement residuals defined in (44), with $z = X_{k-N:k}$. This vector defines the nonlinear least-squares objective, with Jacobian $J = \partial r / \partial z$. The pair $(\hat{X}_{k+1}^{(-)}, P_{k+1}^{(-)})$ then provides the arrival information used to initialize the next step.

Algorithm 1 Moving Horizon Estimation for Slosh Parameter Identification

- 1: Initialize prior mean $\hat{X}_0^{(-)}$ and covariance $P_0^{(-)}$
- 2: **for** $k = N, \dots, T$ **do**
- 3: Extract measurement window $\{y_{k-N}, \dots, y_k\}$
- 4: Build initial guess for $\{X_{k-N:k}\}$
- 5: Solve nonlinear least-squares problem:

$$\min \|r_{\text{arrival}}\|^2 + \|r_{\text{dyn}}\|^2 + \|r_{\text{meas}}\|^2$$

- 6: State Extract optimal augmented states $\hat{X}_{k-N:k}$, where $\hat{X}_k = [\hat{x}_k^T, \hat{\theta}_k^T]^T$
- 7: Compute approximate posterior covariance (Gauss–Newton):

$$P_k^{(+)} \approx (J^T J)^{-1}, \quad J = \frac{\partial r}{\partial z}$$

- 8: Propagate arrival statistics:

$$\hat{X}_{k+1}^{(-)} = A_k \hat{X}_k^{(+)}, \quad P_{k+1}^{(-)} = A_k P_k^{(+)} A_k^T + C_k Q_k C_k^T$$

- 9: **end for**
-

A. Estimator Implementation and Data Processing

The estimation framework developed in the previous sections is applied to CFD-derived measurements of center-of-mass motion and wall force. For each simulated fuel height, the measurement sequence

$$y_k = \begin{bmatrix} q_{\text{meas},k} \\ F_{x,\text{meas},k} \end{bmatrix}$$

is used as input to the moving horizon estimator.

The estimator is implemented over a finite sliding window of length N , where at each time step a batch optimization problem is solved to estimate the augmented state, which includes both the modal states and the time-varying parameter vector. The resulting sequence of parameter estimates,

$$\{\theta_k\}_{k=0}^T,$$

captures the effective reduced-order dynamics required to reproduce the observed force response.

To obtain a compact analytic representation, the estimated parameter trajectories are fit using the structured functions introduced in Section VII, converting discrete estimates into continuous-time functions and enabling construction of smooth coefficient surfaces $\theta(t, h)$ across time and fuel height. The reconstructed force is then obtained by substituting the estimated states and fitted parameter functions into the force model, allowing direct comparison with CFD-derived measurements. This defines a pipeline from raw measurements to an analytic force representation, consisting of parameter estimation, functional approximation, and force reconstruction. The results are organized to evaluate these stages sequentially, isolating their contributions to overall model accuracy.

VIII. RESULTS

A. Overview of Results and Modeling Workflow

The results presented in this section follow the sequence by which the reduced-order force model is constructed and validated. Specifically, the process consists of three stages:

- 1) Identification of time-dependent parametric representations for each coefficient at discrete fuel heights,
- 2) Validation of these analytic representations against the raw estimated parameters,
- 3) Reconstruction of the slosh-induced force using the analytic model and comparison with CFD-derived measurements.

Finally, the validated time-domain models are extended across fuel height to construct a continuous (t, h) -dependent force model suitable for GNC applications.

This organization is important, as the “analytic force model” used in the reconstruction is not assumed *a priori*, but rather constructed from fitted parameter functions $\theta(t, h_i)$ that are themselves identified from CFD data.

B. Parametric Force Model Identification

The slosh-induced wall force is modeled as a linear combination of modal states and input acceleration,

$$F_x(t) = c_1(t, h) q_1(t) + c_2(t, h) \dot{q}_1(t) + c_3(t, h) q_2(t) + c_4(t, h) \dot{q}_2(t) + d(t, h) a_x(t) + b(t, h), \quad (60)$$

where the coefficients $\{c_i, d, b\}$ depend on both time and fuel height to capture transient fluid effects and reduced-order modeling error.

For each discrete fuel height h_i , the coefficient time histories are identified from CFD data using a compact analytic representation of the form

$$\theta(t, h_i) = c_\infty + Ae^{-t/\tau_1} \sin(\omega t + \phi) + Be^{-t/\tau_2}, \quad (61)$$

where $\theta \in \{c_1, c_2, c_3, c_4, d, b\}$ and the parameter set $\{c_\infty, A, \tau_1, \omega, \phi, B, \tau_2\}$ is independently estimated at each height.

This functional form captures the dominant transient and steady-state characteristics observed in the estimated parameters, including oscillatory decay and monotonic convergence.

C. Validation of Analytic Parameter Fits

To verify that the analytic representation in (61) accurately captures the estimated parameter dynamics, the fitted functions are compared against the raw estimated coefficients obtained from the estimator.

The results show that the analytic functions reproduce the dominant features of the parameter trajectories, including transient oscillations, exponential decay rates, and steady-state convergence. Small discrepancies are observed during the early transient, particularly in coefficients associated with velocity-dependent terms (c_2, c_4) and the bias term b . These differences reflect higher-frequency fluid dynamics and nonlinear effects that are not fully captured by the chosen functional form.

Importantly, these discrepancies are localized in time and do not significantly affect the overall structure of the parameter evolution.

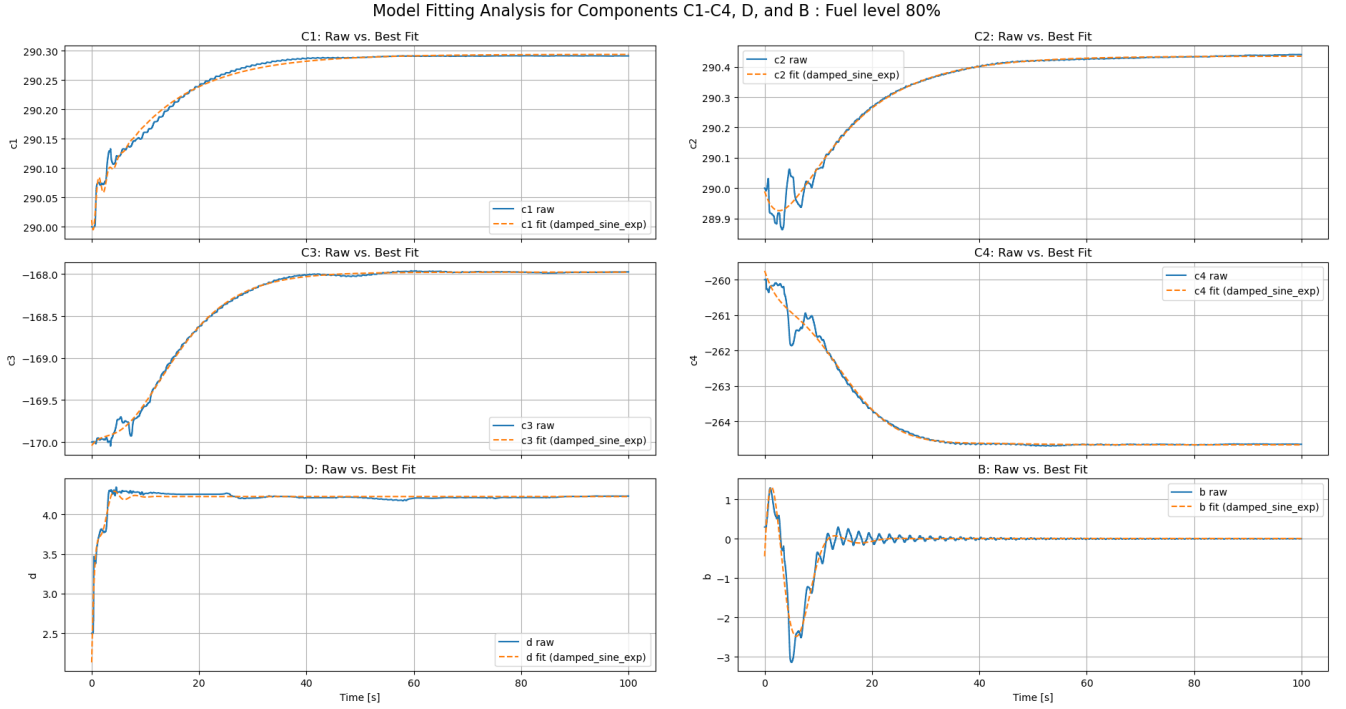


Fig. 4: Comparison between raw estimated parameters and analytic fits for c_1 – c_4 , d , and b at 80% fill level. The analytic functions accurately reproduce the overall trends and decay behavior of the estimated parameters.

D. Force Reconstruction Accuracy

Using the validated analytic parameter functions $\theta(t, h_i)$, the force model in (60) is evaluated to reconstruct the slosh-induced wall force. This reconstruction is compared against the CFD-derived measurements.

Figure 5 shows the force reconstruction for the representative case of 80% fill level, including both the reconstruction obtained using the raw estimated parameters and the fully analytic parametric fits.

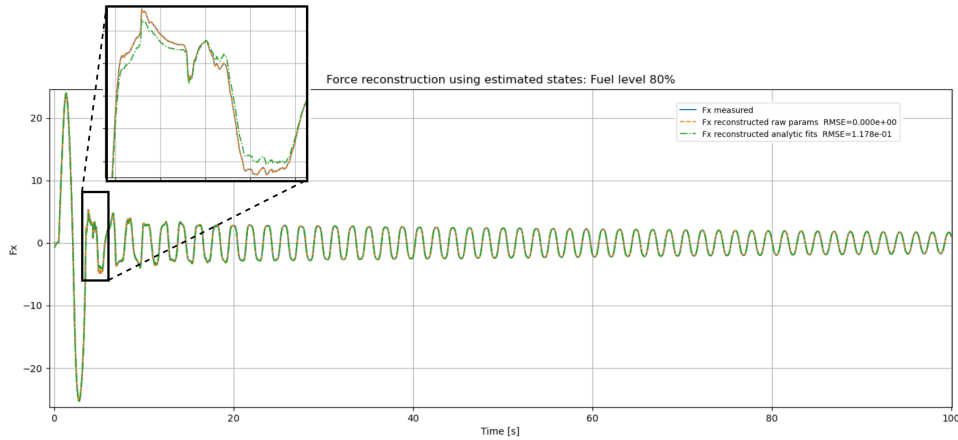


Fig. 5: Wall force reconstruction at 80% fill level. The analytic parametric model closely matches the CFD-derived force, with only minor deviations during the initial transient.

The results demonstrate that the analytic representation preserves the dominant features of the force response across both transient and steady-state regimes. In particular, the oscillatory behavior, decay rate, and steady-state amplitude are accurately captured. The reported root-mean-square error (RMSE) between the analytic reconstruction and the measured force remains small, indicating that the parametric form introduces minimal loss of fidelity relative to the raw parameter estimates.

To further examine the reconstruction quality, the residual error between the measured force and the analytic model is shown in Fig. 6.

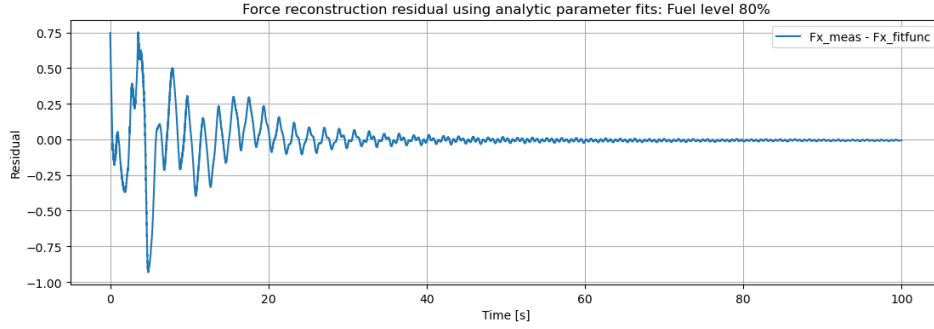


Fig. 6: Residual error between measured and reconstructed force at 80% fill level. The error is largest during the initial transient and rapidly decays to near-zero.

The residual is primarily concentrated in the early-time transient ($t \lesssim 10$ s), where higher-frequency fluid effects and nonlinearities are most pronounced. Beyond this region, the error rapidly decays and remains close to zero, indicating that the reduced-order parametric model captures the dominant long-term dynamics with high accuracy.

Importantly, these results demonstrate that the force reconstruction is robust to small discrepancies in individual coefficient fits. Even though minor parameter-level errors exist, the overall force prediction remains highly accurate due to the preserved structure of the model.

Finally, it is noted that similar reconstruction accuracy is observed across all fuel heights considered, indicating that the parametric structure generalizes well with respect to fill level.

E. Extension to Continuous (t, h) Force Model

Having validated the analytic time-domain representations $\theta(t, h_i)$ at discrete fuel heights, the next step is to construct continuous coefficient surfaces over both time and fuel height.

The full coefficient surfaces are constructed using a *curve-first interpolation strategy*. This approach proceeds in two stages:

- 1) For each height h_i , an analytic time-domain model $\theta(t, h_i)$ is identified using (61).
- 2) For each fixed time t_k , the discrete values $\theta(t_k, h_i)$ are interpolated across fuel height to define $\theta(t, h)$.

This yields the representation

$$\theta(t, h) = \mathcal{I}_h \{ \theta(t, h_i) \}, \quad (62)$$

In this work, the interpolation operator $\mathcal{I}_h$ is implemented using a shape-preserving piecewise cubic Hermite interpolant (PCHIP). On each interval $[h_i, h_{i+1}]$, the function $\theta(t_k, h)$ is represented by a cubic polynomial constructed from the nodal values $\theta(t_k, h_i)$ together with locally estimated slopes $\partial\theta/\partial h$. This choice ensures that the interpolation preserves the monotonicity and overall shape of the data, thereby avoiding the spurious oscillations commonly associated with standard cubic splines. Such behavior is particularly important in the present application, as non-physical oscillations in the interpolated force coefficients could introduce artificial dynamics and degrade control performance during fuel height transitions.

This construction ensures that:

- the identified time-domain dynamics at each simulated height are preserved exactly,
- smooth variation with respect to fuel height is obtained,
- non-physical oscillations associated with global polynomial or spline fits are avoided.

The resulting coefficient surfaces $\theta(t, h)$ are shown in Fig. 7, demonstrating consistent behavior across both time and fill level.

Substituting (61) and (62) into (60) yields a reduced-order force model that is analytic in time and smoothly varying in fuel height, while retaining fidelity to the underlying CFD data.

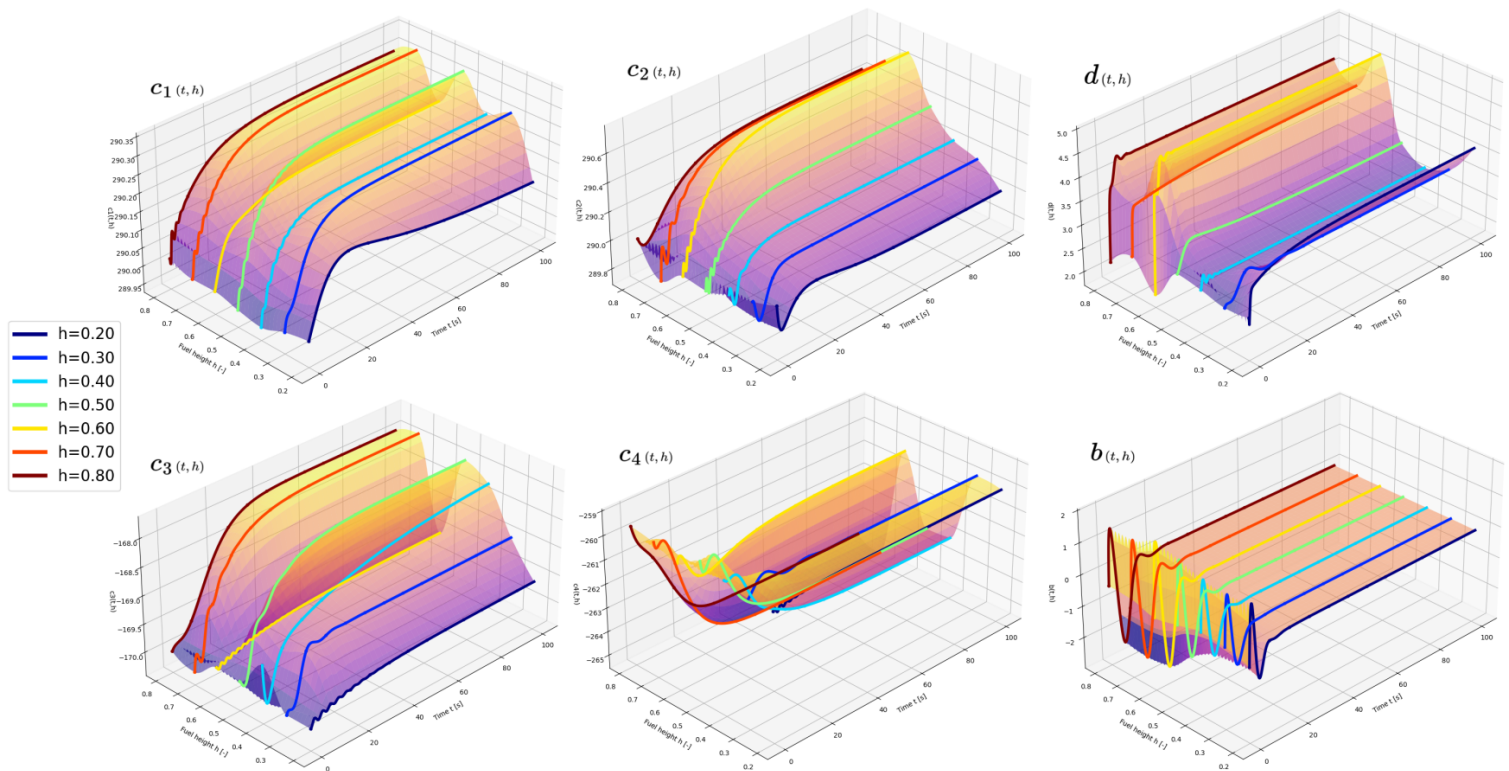


Fig. 7: Identified coefficient surfaces as functions of time and fuel height.

This representation provides a compact and computationally efficient model suitable for GNC simulations, while preserving the dominant transient and steady-state characteristics of slosh-induced forces.

IX. CONCLUSION

A reduced-order slosh estimation framework has been developed for a laterally excited partially filled tank using CFD-derived center-of-mass motion and wall force measurements. A two-mode model captures the dominant slosh dynamics while remaining suitable for recursive estimation, and a finite-horizon MAP formulation of MHE yields a covariance-weighted cost with measurement, arrival, and dynamic-consistency terms. A local second-order approximation provides a curvature-based posterior covariance together with a recursive arrival covariance propagation law, resulting in a consistent framework linking CFD data, reduced-order dynamics, and joint state-parameter estimation.

The identified parametric force model achieves high reconstruction accuracy across both transient and steady-state regimes, with residual errors rapidly decaying after the initial transient. These results demonstrate that high-fidelity slosh dynamics can be systematically compressed into a low-dimensional, time-varying parametric representation without sacrificing force prediction accuracy. Notably, the analytic parameterization preserves force-level fidelity despite minor coefficient-level discrepancies, highlighting the robustness of the approach. This establishes a viable pathway for incorporating CFD-informed slosh effects into real-time GNC simulation and control design frameworks. Future work will focus on integrating these continuous parametric models into spacecraft flight simulation environments to evaluate the impact of slosh-induced forces on closed-loop attitude control performance and pointing accuracy.

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